智能健康设备养老院场景可用性测试知情同意书

致：至爱老年医疗护理院及尊敬的参与者：

您好！我们是来自吉林大学公共卫生学院（JLU-SPH）的学生产品研发团队。我们正在参加一项名为国际基因工程机器大赛（iGEM）的非营利性学术项目，致力于开发一款面向老年用户的智能健康监测设备，力求让它对视力、听力衰退的长辈们更加友好、易用。

为优化产品设计，我们正在开展用户调研。我们计划于 2026 年 4 月 10 日，在贵院开展一次设备原型测试与访谈。为确保您/院方在充分了解所有细节的基础上自愿参与，特此作如下详细说明：

一、测试与调研内容概述

1. 我们将做什么：

- **体验设备：** 邀请您体验我们最新版的设备原型机（例如，一款带有推拉滑盖的测试设备）。
- **自然操作：** 请您在自然的状态下，尝试操作该设备（如打开滑盖、查看屏幕颜色等）。
- **轻松交流：** 我们的研究员会与您进行约 20-40 分钟的对话，询问您在使用过程中的真实感受、困惑或抱怨。
- **全程记录：** 我们会观察并记录您与设备的互动过程。

2. 您的参与方式：

- 您只需像平时使用日常物品那样操作，**这里没有“正确”或“错误”的操作，只有机器设计得“好不好”**。
- 您可以随时提问或表达困难（例如“这个我看不清”、“太费劲了”），您的反馈对我们至关重要。

二、隐私、保密与数据使用承诺

为准确记录您的宝贵反馈，我们希望在测试中进行录音和录像/拍照，但我们将严格保护您的隐私：

1. **记录内容：** 录音仅用于记录访谈内容；录像/拍照仅用于分析手部动作、交互方式及表情（判断是否困惑）。
2. **匿名化处理：** 在任何对外的报告、学术海报或我们的国际比赛公开网站（iGEM Wiki）上，您的真实姓名、及其他个人身份信息都将被**隐去或脱敏打码处理**，代之以化名（如“李奶奶”）。

3. **数据存储与销毁：** 原始录音录像文件将加密保存在团队专属硬盘中，仅限核心研究人员查看。所有原始影像数据将在本次比赛周期结束后（预计为 2026 年 12 月）予以彻底永久销毁。绝不会用于任何商业营销或提供给第三方。
4. **选择权：** 您有权拒绝录音或录像。如果您不同意，测试仍可进行，我们将仅采用纸笔记录。

三、风险、补偿与重要免责声明

1. **非医疗诊断声明（重要）：** 本次测试的设备仅为**处于研发阶段的原型机，并非成熟的医疗器械**。本次活动纯粹为了“改进设备外观和操作设计”，**不提供任何形式的健康检查、疾病诊断或医疗建议**。您的日常护理与医疗服务绝不会受到任何影响。
2. **补偿与随时终止权：** 参与测试完全免费。如果在测试中您因任何原因感到不适、疲惫或不愿继续，请随时告诉我们，测试将立即无条件停止。为感谢您的时间和贡献，我们将为您准备一份小礼品。

四、自愿参与及退出机制

您的参与是**完全自愿**的。即使您已签署本同意书，也有权在测试开始前或过程中的任何时刻无理由退出。*补充说明：* 如果您在测试结束后希望撤回您的记录数据，请在 2026 年 6 月 10 日前联系我们，我们将立刻删除。在此日期之后，由于数据可能已匿名化并混入整体学术报告中，我们将无法单独分离并删除您的信息。

五、联系与咨询方式

如您或您的家属对本次活动有任何疑问、建议，或希望退出测试，请随时联系我们：

- **日常对接与测试安排（学生负责人）：尹桐辉**，电话：18143095975
- **项目学术与伦理责任人（指导老师）：** [姓名] 教授，办公电话：[号码]，邮箱：[邮箱地址]

请仔细阅读（或由工作人员为您宣读）以上全部内容。如您已完全了解并同意，请在下方签名确认。

养老院（院方）确认

我方，至爱老年医疗护理院，已充分知悉并理解本次测试的全部内容及免责声明。我方同意 JLU-SPH 团队在指定日期于我院开展此项活动，并将协助在确保长辈身心安全的前提下进行自愿招募与沟通。

机构盖章/负责人签字：_____

日期：2026 年 ___ 月 ___ 日

测试参与者（或其授权代理人）

确认我已阅读（或已由工作人员向我清晰宣读并解释）本同意书。我理解本次活动的性质是产品设计调研，不提供医疗诊断。我了解将被记录的方式及我享有的隐私保护和退出权利。我自愿参加。

关于记录方式的同意（请勾选）：

● 我同意进行录音。

□是　□否

● 我同意进行录像/拍照（面部会打码脱敏）。

□是　□否

参与者签名：_____

日期：2026 年 ___ 月 ___ 日

（如参与者因视力、书写困难等原因无法亲自阅读或签署，或经院方评估需代理人同意，请填写以下部分）

代理人声明：本人作为代理人（家属 / 法定监护人 / 院方指定负责人），已确保其知晓并理解本同意书全部内容，并在尊重其本人意愿的前提下，代为同意参与本次测试，并同意上述勾选的记录方式。

参与者姓名：_____

代理人签名：_____

与参与者关系：_____

日期：2026 年 ___ 月 ___ 日

Informed Consent Form for Usability Testing of

Smart Health Devices in Nursing Home Scenarios

To: Zhi'ai Elderly Medical and Nursing Home and Respected Participants:

Hello! We are the student product research and development team from the School of Public Health, Jilin University (JLU-SPH). We are participating in a non-profit academic project called the International Genetically Engineered Machine (iGEM) competition. We are developing a smart health monitoring device aimed at elderly users, striving to make it more user-friendly and accessible for seniors with declining vision and hearing.

To optimize our product design, we are conducting user research. We plan to conduct a device prototype test and interview at your facility on April 10, 2026. To ensure that you/the institution voluntarily participate with a full understanding of all details, we hereby provide the following detailed explanation:

I. Overview of Testing and Research Content

1. What we will do:

- **Experience the Device:** Invite you to experience our latest device prototype (e.g., a test device with a push-pull sliding cover).
- **Natural Operation:** Ask you to attempt to operate the device in a natural state (e.g., opening the cover, checking screen colors).
- **Casual Conversation:** Our researchers will have a 20-40 minute conversation with you to ask about your true feelings, confusions, or complaints during the use process.
- **Whole-Process Recording:** We will observe and record your interaction with the device.

2. How you participate:

- You only need to operate it as you would an everyday item. **There are no "right" or "wrong" operations here, only whether the machine is designed "well" or "poorly".**
- You can ask questions or express difficulties at any time (e.g., "I can't see this clearly", "This takes too much effort"). Your feedback is crucial to us.

II. Commitments to Privacy, Confidentiality, and Data Use

To accurately record your valuable feedback, we wish to conduct audio and video/photo recording during the test, but we will strictly protect your privacy:

1. **Recorded Content:** Audio is only used to record interview content; video/photos are only used to analyze hand movements, interaction methods, and facial expressions (to determine if there is confusion).
2. **Anonymization:** In any external reports, academic posters, or our international competition's public website (iGEM Wiki), your real name, facial features, and other personal identification information will be **concealed or pixelated/desensitized**, and replaced with a pseudonym (e.g., "Grandma Li").
3. **Data Storage and Destruction:** Original audio and video files will be encrypted and stored on the team's dedicated hard drive, accessible only to core researchers. All original image data will be permanently and thoroughly destroyed after the current competition cycle ends (expected December 2026). Data will never be used for any commercial marketing or provided to third parties.
4. **Right to Choose:** You have the right to refuse audio or video recording. If you do not consent, the test can still proceed, and we will rely solely on written notes.

III. Risks, Compensation, and Important Disclaimers

1. **Non-Medical Diagnosis Statement (Important):** The device used in this test is **only a prototype in the research and development stage, not a mature medical device**. This activity is purely for "improving the appearance and operational design of the device," and **does not provide any form of health check, disease diagnosis, or medical advice**. Your daily care and medical services will not be affected in any way.
2. **Compensation and Right to Terminate:** Participation in the test is completely free. If, for any reason, you feel unwell, exhausted, or unwilling to continue during the test, please tell us at any time, and the test will be stopped immediately and unconditionally. To thank you for your time and contribution, we will prepare a small gift for you.

III. Voluntary Participation and Withdrawal Mechanism

Your participation is **completely voluntary**. Even if you have signed this consent form, you have the right to withdraw without giving any reason at any time before or during the test.*Additional Note:* If you wish to withdraw your recorded data after the test ends, please contact us before June 10th, 2026, and we will delete it immediately. After this date, because the data may have been anonymized and integrated into the overall academic report, we will be unable to separate and delete your information individually.

IV. Contact and Consultation Information

If you or your family members have any questions, suggestions, or wish to withdraw from the test, please contact us at any time:

- **Daily Coordination and Test Arrangements (Student Lead):** Tonghui Yin , Phone: 18143095975
- **Project Academic and Ethical Supervisor (Principal Investigator):** Professor [Name], Office Phone: [Number], Email: [Email Address]

Please read carefully (or have a staff member clearly read to you) all the above contents. If you fully understand and agree, please sign below to confirm.

Nursing Home (Institution) Confirmation

We, Zhi'ai Elderly Medical and Nursing Home, are fully aware of and understand the entire content of this test and the disclaimer. We agree that Team JLU-SPH will conduct this activity at our institution on the specified date, and we will assist in voluntary recruitment and communication under the premise of ensuring the physical and mental safety of the elders.

Institution Seal/Signature of Person in Charge: _____

Date: ___ / ___ / 2026

Test Participant (or Authorized Proxy) Confirmation

I have read (or a staff member has clearly read and explained to me) this consent form. I understand that the nature of this activity is product design research and does not provide medical diagnosis. I am aware of how I will be recorded and the privacy protection and withdrawal rights I enjoy. I voluntarily participate.

Consent Regarding Recording Methods (Please check):

● I consent to audio recording.

□ Yes □ No

● I consent to video/photo recording (facial features will be pixelated/desensitized).

□ Yes □ No

Participant Signature: _____

Date: ___ / ___ / 2026

(If the participant cannot personally read or sign due to declining vision, writing difficulties, etc., or if the institution assesses that a proxy's consent is required, please fill out the following section)

Proxy Declaration: As the proxy (family member / legal guardian / institutional designated representative), I have ensured that they know and understand the entire content of this consent form, and on the premise of respecting their own wishes, I consent on their behalf to participate in this test, and agree to the recording methods checked above.

Participant Name: _____

Proxy Signature: _____

Relationship to Participant: _____

Date: ___ / ___ / 2026